Foundation University Islamabad

School of Science and Technology

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Subject: Oops Lab **Section:** B

LAB REPORT NO.2

Exercises: LAB 02

TASK 1:

Write a class Student with three data members: RollNo, Name, and Marks. The class should have:

i. A default constructor to initialize values to 0 or empty.
ii. A parameterized constructor to initialize values using arguments.
iii. A display() function to print the details.
iv. In main(), create two objects: one with default values and another with provided values.

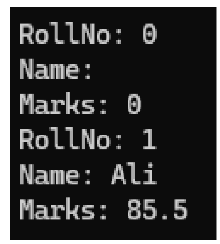
SOURCE CODE:

```
#include <iostream>
using namespace std;

class Student {
public:
   int RollNo;
   string Name;
   float Marks;

   Student() {
      RollNo = 0;
      Name = "";
      Marks = 0;
   }

   Student(int r, string n, float m) {
      RollNo = r;
      Name = n;
      Marks = m;
   }

   void display() {
      cout << "RollNo: " << RollNo << endl;
      cout << "Name: " << Name << endl;
```

```
        cout << "Marks: " << Marks << endl;
    }
};

int main() {
    Student s1;
    Student s2(1, "Ali", 85.5);

    s1.display();
    s2.display();

    return 0;
}
```

OUTPUT:

RollNo: 0
Name:
Marks: 0
RollNo: 1
Name: Ali
Marks: 85.5



TASK 2:

Write a class Rectangle with two private data members: length and width. The class should have:

i. A parameterized constructor to initialize values.

ii. A copy constructor that initializes a new object with values from another object.

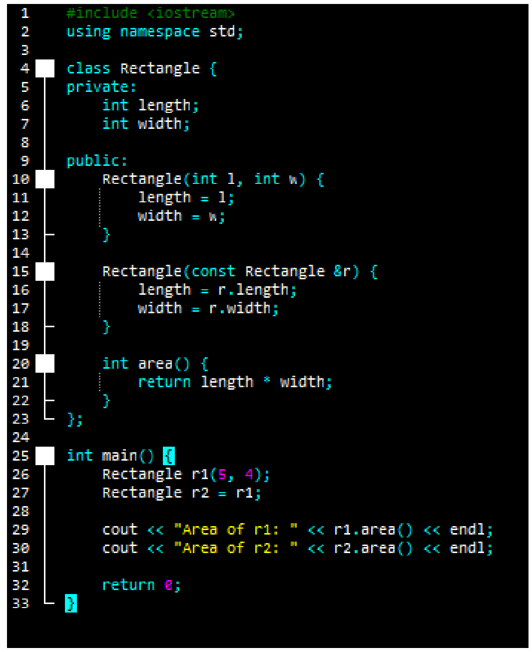
iii. An area() function to return the area.

iv. In main(), create an object and then create a second object using the copy constructor, displaying both areas.

SOURCE CODE:

```
#include <iostream>
using namespace std;

class Rectangle {
private:
    int length;
    int width;

public:
    Rectangle(int l, int w) {
        length = l;
        width = w;
    }

    Rectangle(const Rectangle &r) {
```

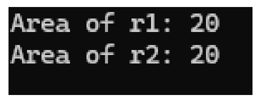
```
        length = r.length;
        width = r.width;
    }

    int area() {
        return length * width;
    }
};

int main() {
    Rectangle r1(5, 4);
    Rectangle r2 = r1;

    cout << "Area of r1: " << r1.area() << endl;
    cout << "Area of r2: " << r2.area() << endl;

    return 0;
}
```



```
#include <iostream>
using namespace std;

class Rectangle {
private:
    int length;
    int width;

public:
    Rectangle(int l, int w) {
        length = l;
        width = w;
    }

    Rectangle(const Rectangle &r) {
        length = r.length;
        width = r.width;
    }

    int area() {
        return length * width;
    }
};

int main() {
    Rectangle r1(5, 4);
    Rectangle r2 = r1;

    cout << "Area of r1: " << r1.area() << endl;
    cout << "Area of r2: " << r2.area() << endl;

    return 0;
}
```

OUTPUT:

Area of r1: 20
Area of r2: 20



```
Area of r1: 20
Area of r2: 20
```

TASK 3:

Write a class Car with three private data members: Model, Company, and Price. The class should have:

i. A parameterized constructor to initialize values.

ii. A destructor that prints "Object Destroyed" when an object is deleted.

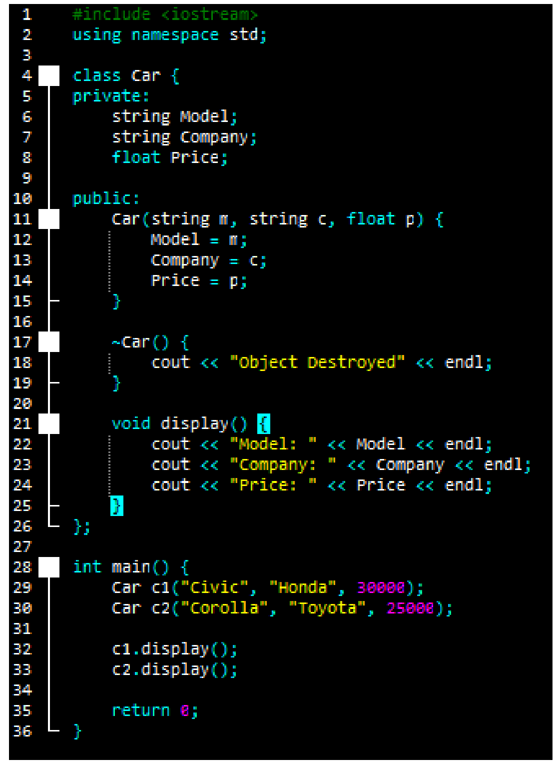
iii. A display() function to print details.

iv. In main(), create two objects and observe when the destructor is called.

SOURCE CODE:

```
#include <iostream>
using namespace std;

class Car {
private:
    string Model;
    string Company;
    float Price;

public:
    Car(string m, string c, float p) {
        Model = m;
        Company = c;
        Price = p;
    }
```

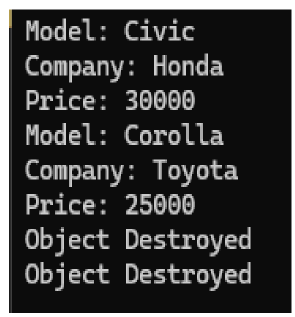
```
   ~Car() {
      cout << "Object Destroyed" << endl;
   }

   void display() {
      cout << "Model: " << Model << endl;
      cout << "Company: " << Company << endl;
      cout << "Price: " << Price << endl;
   }
};

int main() {
   Car c1("Civic", "Honda", 30000);
   Car c2("Corolla", "Toyota", 25000);

   c1.display();
   c2.display();

   return 0;
}
```



```
#include <iostream>
using namespace std;

class Car {
private:
    string Model;
    string Company;
    float Price;

public:
    Car(string m, string c, float p) {
        Model = m;
        Company = c;
        Price = p;
    }

    ~Car() {
        cout << "Object Destroyed" << endl;
    }

    void display() {
        cout << "Model: " << Model << endl;
        cout << "Company: " << Company << endl;
        cout << "Price: " << Price << endl;
    }
};

int main() {
    Car c1("Civic", "Honda", 30000);
    Car c2("Corolla", "Toyota", 25000);

    c1.display();
    c2.display();

    return 0;
}
```

<u>OUTPUT:</u>

Model: Civic
Company: Honda
Price: 30000
Model: Corolla
Company: Toyota
Price: 25000
Object Destroyed
Object Destroyed



```
Model: Civic
Company: Honda
Price: 30000
Model: Corolla
Company: Toyota
Price: 25000
Object Destroyed
Object Destroyed
```

<u>TASK 4:</u>

Write a class Circle with one private data member: radius. The class should have:

i. A default constructor that initializes radius = 1.

ii. A parameterized constructor that sets radius based on an argument.

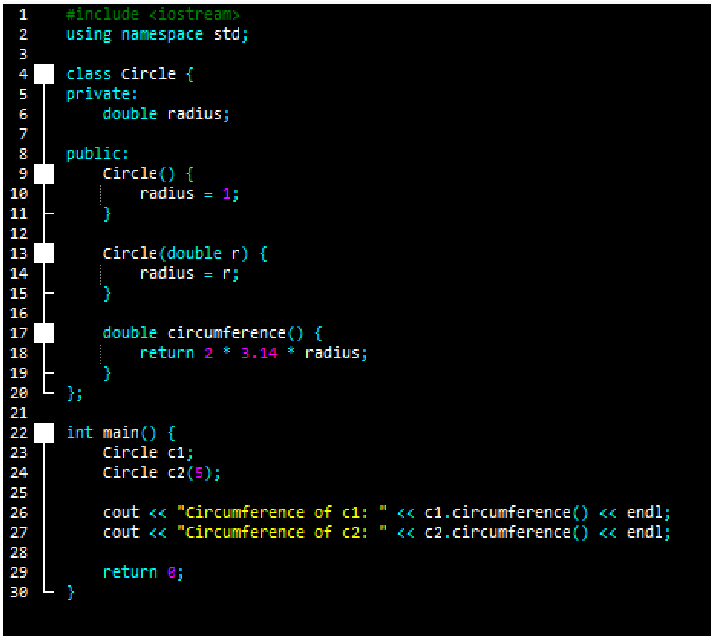
iii. A circumference() function that returns the circumference.

iv. In main(), create two objects using both constructors and display their circumferences.

<u>SOURCE CODE:</u>

```
#include <iostream>
using namespace std;

class Circle {
private:
   double radius;
```

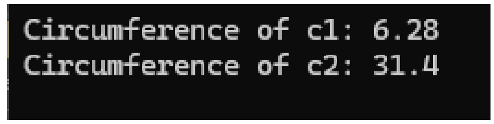
```
public:
   Circle() {
      radius = 1;
   }

   Circle(double r) {
      radius = r;
   }

   double circumference() {
      return 2 * 3.14 * radius;
   }
};

int main() {
   Circle c1;
   Circle c2(5);

   cout << "Circumference of c1: " << c1.circumference() << endl;
   cout << "Circumference of c2: " << c2.circumference() << endl;

   return 0;
}
```



```
#include <iostream>
using namespace std;

class Circle {
private:
    double radius;

public:
    Circle() {
        radius = 1;
    }

    Circle(double r) {
        radius = r;
    }

    double circumference() {
        return 2 * 3.14 * radius;
    }
};

int main() {
    Circle c1;
    Circle c2(5);

    cout << "Circumference of c1: " << c1.circumference() << endl;
    cout << "Circumference of c2: " << c2.circumference() << endl;

    return 0;
}
```

OUTPUT:

Circumference of c1: 6.28
Circumference of c2: 31.4



TASK 5:

Write a class Resource with a private data member value. The class should have:

i. A parameterized constructor to initialize value.

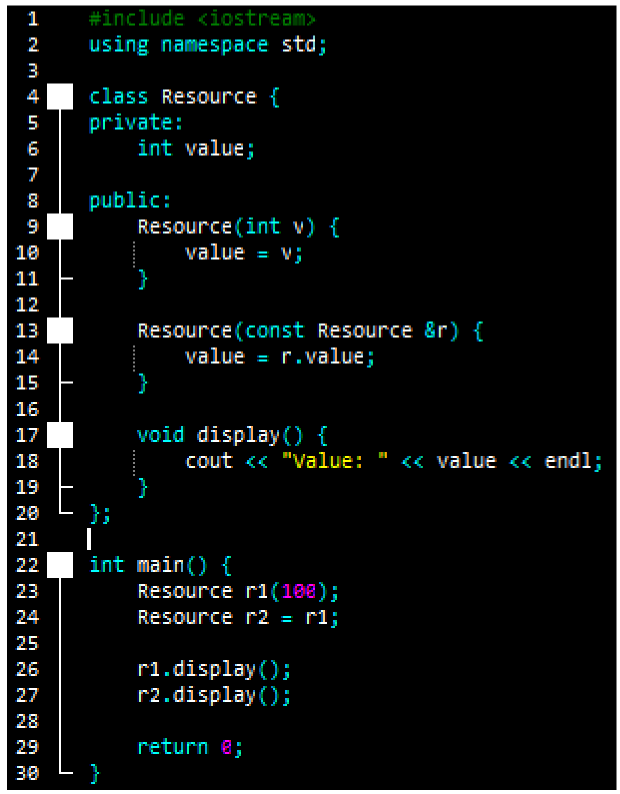
ii. A move constructor to transfer ownership.

iii. A display() function to print the value.

iv. In main(), create an object and then use move() to initialize another object, displaying both values.

SOURCE CODE:

```
#include <iostream>
using namespace std;

class Resource {
private:
    int value;

public:
    Resource(int v) {
        value = v;
    }

    Resource(const Resource &r) {
        value = r.value;
    }

    void display() {
        cout << "Value: " << value << endl;
    }
};
int main() {
    Resource r1(100);
    Resource r2 = r1;

    r1.display();
    r2.display();

    return 0;
}
```



OUTPUT:

Value: 100
Value: 100